# COMPTIA N+

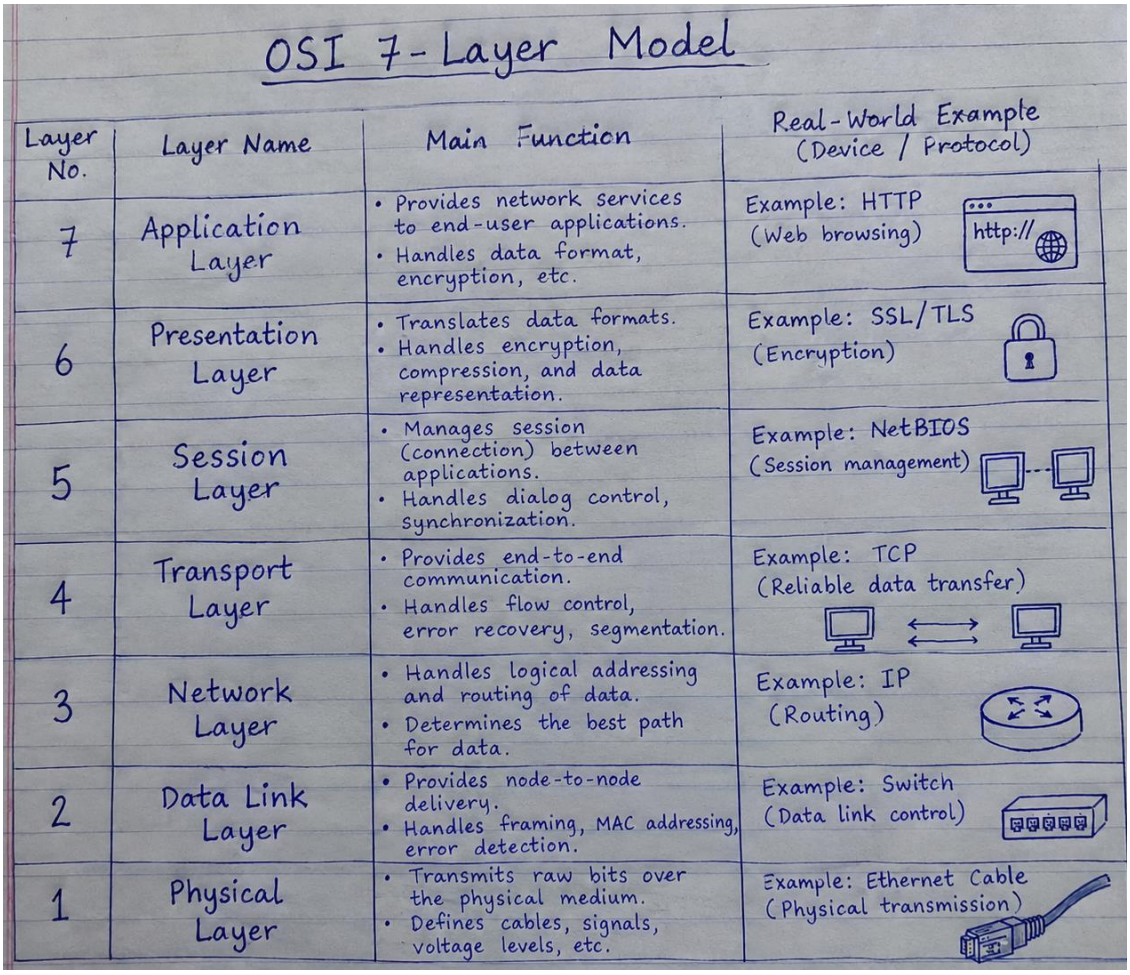
## N+ - Network Fundamental and building networks

# OSI LAYER

# TASKS:-1

Draw a diagram of the OSI 7-layer model and label each layer with its correct name and a real-world example of a device or protocol that operates at that layer (e.g., Switch at Data Link, HTTP at Application)

# RESULT



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# TASKS:-2

Match the following networking scenarios to the correct OSI layer:

a) Encrypting data before sending,

b) Routing packets across networks,

c) Converting data to electrical signals,

d) Opening a web page in your browser,

e) Detecting and correcting errors in frames.

Write your answers as 'Scenario: Layer Name (Layer Number)'.

## RESULT

a) **Encrypting data before sending:**

Presentation Layer (Layer 6)

b) **Routing packets across networks:**

Network Layer (Layer 3)

c) **Converting data to electrical signals:**

Physical Layer (Layer 1)

d) **Opening a web page in your browser:**

Application Layer (Layer 7)

e) **Detecting and correcting errors in frames:**

Data Link Layer (Layer 2)

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# TASKS:-3

Pick any app you use daily (like WhatsApp, Instagram, or Zomato) and describe how at least three different OSI layers are involved when you send or receive data in that app.

# RESULT

**Example: WhatsApp**

When I send or receive a message on WhatsApp, different OSI layers work together:

1. **Application Layer (Layer 7):**
   WhatsApp uses application-level protocols and services to send messages, photos, and other data between users.
2. **Transport Layer (Layer 4):**
   This layer helps provide communication between my phone and the WhatsApp server and manages the delivery of data.
3. **Network Layer (Layer 3):**
   IP addressing and routing are used to move the data packets from my phone through the Internet to the WhatsApp server and then to the receiver.
4. **Data Link Layer (Layer 2):**
   It transfers data between my phone and the Wi-Fi router using frames and MAC addresses.
5. **Physical Layer (Layer 1):**
   It transmits the data as signals through Wi-Fi, mobile network, or cables used in the network.

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# TASKS:-4

Given this scenario: You are unable to access a website, but you can successfully ping its IP address. Identify which OSI layer(s) might be causing the problem and explain your reasoning.

# RESULT

## Scenario: Website is Not Opening but Ping is Successful

If I am unable to open a website, but I can successfully ping its IP address, it means that the basic network connectivity is working.

The problem may mainly be related to the **Transport Layer (Layer 4)** or **Application Layer (Layer 7)**.

- **Transport Layer (Layer 4):** The TCP connection may be blocked, or the required port such as **80 (HTTP)** or **443 (HTTPS)** may not be accessible.
- **Application Layer (Layer 7):** The website's HTTP/HTTPS service may be down, unavailable, or incorrectly configured.

**Reason:** Ping uses ICMP to check whether the destination IP is reachable. Since ping is successful, basic IP connectivity is working. However, opening a website requires a working TCP connection and an HTTP/HTTPS service. Therefore, I would check the Transport and Application layers.